XLSX Creation, Editing, and Analysis

Overview

This skill covers working with Excel files using Python libraries: **openpyxl** for Excel-specific features (formatting, formulas, pivot tables) and **pandas** for data analysis.

Reading Data

With pandas (recommended for analysis)
```python
import pandas as pd

df = pd.read_excel('file.xlsx')  # First sheet
all_sheets = pd.read_excel('file.xlsx', sheet_name=None)  # All sheets as dict
df.head()      # Preview
df.describe()  # Statistics
```

With openpyxl (for cell-level access)
```python
from openpyxl import load_workbook

wb = load_workbook('file.xlsx')
ws = wb.active
value = ws['A1'].value

# Read calculated values (not formulas)
wb = load_workbook('file.xlsx', data_only=True)
```

Creating Excel Files

```python
from openpyxl import Workbook
from openpyxl.styles import Font, PatternFill, Alignment

wb = Workbook()
ws = wb.active
ws.title = "Data"

# Add data
ws['A1'] = 'Header'
ws.append(['Row', 'of', 'data'])

# Formatting
ws['A1'].font = Font(bold=True)
ws['A1'].fill = PatternFill('solid', start_color='2c3e50')

wb.save('output.xlsx')
```

Creating Pivot Tables

Pivot tables summarize data by grouping and aggregating. Use openpyxl's pivot table API.

CRITICAL: All pivot tables MUST use `cacheId=0`. Using any other cacheId (1, 2, etc.) will cause openpyxl to fail when reading the file back with `KeyError`. This is an openpyxl limitation.

Basic Pivot Table Structure

```python
from openpyxl import Workbook
from openpyxl.pivot.table import TableDefinition, Location, PivotField, DataField, RowColField
from openpyxl.pivot.cache import CacheDefinition, CacheField, CacheSource, WorksheetSource, SharedItems
```

```
# 1. Create workbook with source data
wb = Workbook()
data_ws = wb.active
data_ws.title = "SourceData"

# Write your data (with headers in row 1)
data = [
    ["CategoryName", "ProductName", "Quantity", "Revenue"],
    ["Beverages", "Chai", 25, 450.00],
    ["Seafood", "Ikura", 12, 372.00],
    # ... more rows
]
for row in data:
    data_ws.append(row)

num_rows = len(data)

# 2. Create pivot table sheet
pivot_ws = wb.create_sheet("PivotAnalysis")

# 3. Define the cache (source data reference)
cache = CacheDefinition(
    cacheSource=CacheSource(
        type="worksheet",
        worksheetSource=WorksheetSource(
            ref=f"A1:D{num_rows}",  # Adjust to your data range
            sheet="SourceData"
        )
    ),
    cacheFields=[
        CacheField(name="CategoryName", sharedItems=SharedItems(count=8)),
        CacheField(name="ProductName", sharedItems=SharedItems(count=40)),
        CacheField(name="Quantity", sharedItems=SharedItems()),
        CacheField(name="Revenue", sharedItems=SharedItems()),
    ]
)

# 4. Create pivot table definition
pivot = TableDefinition(
    name="RevenueByCategory",
    cacheId=0,  # MUST be 0 for ALL pivot tables - any other value breaks openpyxl
    dataCaption="Values",
    location=Location(
        ref="A3:B10",  # Where pivot table will appear
        firstHeaderRow=1,
        firstDataRow=1,
        firstDataCol=1
    ),
)

# 5. Configure pivot fields (one for each source column)
# Fields are indexed 0, 1, 2, 3 matching cache field order

# Field 0: CategoryName - use as ROW
pivot.pivotFields.append(PivotField(axis="axisRow", showAll=False))
# Field 1: ProductName - not used (just include it)
pivot.pivotFields.append(PivotField(showAll=False))
# Field 2: Quantity - not used
pivot.pivotFields.append(PivotField(showAll=False))
# Field 3: Revenue - use as DATA (for aggregation)
pivot.pivotFields.append(PivotField(dataField=True, showAll=False))

# 6. Add row field reference (index of the field to use as rows)
pivot.rowFields.append(RowColField(x=0))  # CategoryName is field 0
```

```python
# 7. Add data field with aggregation
pivot.dataFields.append(DataField(
    name="Total Revenue",
    fld=3,  # Revenue is field index 3
    subtotal="sum"  # Options: sum, count, average, max, min, product, stdDev, var
))

# 8. Attach cache and add to worksheet
pivot.cache = cache
pivot_ws._pivots.append(pivot)

wb.save('output_with_pivot.xlsx')
```

Common Pivot Table Configurations

Count by Category (Order Count)
```python
# Row field: CategoryName (index 0)
# Data field: Count any column

pivot.rowFields.append(RowColField(x=0))  # CategoryName as rows
pivot.dataFields.append(DataField(
    name="Order Count",
    fld=1,  # Any field works for count
    subtotal="count"
))
```

Sum by Category (Total Revenue)
```python
pivot.rowFields.append(RowColField(x=0))  # CategoryName as rows
pivot.dataFields.append(DataField(
    name="Total Revenue",
    fld=3,  # Revenue field
    subtotal="sum"
))
```

Two-Dimensional Pivot (Rows and Columns)
```python
# CategoryName as rows, Quarter as columns, revenue as values
pivot.pivotFields[0] = PivotField(axis="axisRow", showAll=False)  # Category = row
pivot.pivotFields[1] = PivotField(axis="axisCol", showAll=False)  # Quarter = col
pivot.pivotFields[3] = PivotField(dataField=True, showAll=False)  # Revenue = data

pivot.rowFields.append(RowColField(x=0))  # Category
pivot.colFields.append(RowColField(x=1))  # Quarter
pivot.dataFields.append(DataField(name="Revenue", fld=3, subtotal="sum"))
```

Multiple Data Fields
```python
# Show both count and sum
pivot.dataFields.append(DataField(name="Order Count", fld=2, subtotal="count"))
pivot.dataFields.append(DataField(name="Total Revenue", fld=3, subtotal="sum"))
```

Pivot Table Field Configuration Reference

axis value	Meaning
`"axisRow"`	Field appears as row labels
`"axisCol"`	Field appears as column labels
`"axisPage"`	Field is a filter/slicer
(none)	Field not used for grouping

subtotal value	Aggregation
`"sum"`	Sum of values
`"count"`	Count of items
`"average"`	Average/mean
`"max"`	Maximum value
`"min"`	Minimum value
`"product"`	Product of values
`"stdDev"`	Standard deviation
`"var"`	Variance

Important Notes

1. **CRITICAL - cacheId must be 0**: ALL pivot tables must use `cacheId=0`. Using 1, 2, etc. will ca
use `KeyError` when reading the file
2. **Field indices must match**: The order of `cacheFields` must match your source data columns
3. **Location ref**: The `ref` in Location is approximate - Excel will adjust when opened
4. **Cache field count**: For categorical fields, set `count` parameter to approximate number of uni
que values
5. **Multiple pivots**: Each pivot table needs its own sheet for clarity
6. **Values populate on open**: Pivot table values are calculated when the file is opened in Excel/L
ibreOffice, not when created

Working with Existing Excel Files

```python
from openpyxl import load_workbook

wb = load_workbook('existing.xlsx')
ws = wb['SheetName']

# Modify
ws['A1'] = 'New Value'
ws.insert_rows(2)

# Add new sheet
new_ws = wb.create_sheet('Analysis')

wb.save('modified.xlsx')
```

Best Practices

1. **Use pandas for data manipulation**, openpyxl for Excel features
2. **Match field indices carefully** when creating pivot tables
3. **Test with small data first** before scaling up
4. **Name your sheets clearly** (e.g., "SourceData", "RevenueByCategory")
5. **Document your pivot table structure** with comments in code